Institutional quality, climate change impacts, and regional economic growth

Evidence from Peru

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- Contribution
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Motivation

- Regional economic growth is a central topic in economic geography, as scholars seek to understand the factors that drive prosperity and regional disparities (Hansen, 2022). Traditional theories emphasize the role of:
  - physical capital (Solow, 1956; Swan, 1956)
  - human capital and innovation (Romer, 1986; Lucas, 1988)

- However, these explanations have become insufficient to account for persistent regional inequalities and two bodies of literature have emerged in parallel to contribute to the discussion: institutional quality and climate change

- Institutional quality: institutions influence economic performance through multiple channels, including productivity, innovation, entrepreneurship, and resource allocation efficiency (Rodríguez-Pose, 2013; Rodríguez-Pose and Ketterer, 2020)

- Climate change: temperature anomalies and extreme weather events, reduce labor productivity, damage agricultural output, and deter capital investment, thereby widening regional disparities (Burke et al., 2015; Dell et al., 2012; Kalkuhl and Wenz, 2020)

Motivation

This paper

- Institutional efficiency is not merely a determinant of regional growth, but also it is a fundamental moderator of how regional economies absorb and respond to climate variability

- Regions have different economic structures and institutional capacities, absorbing climate disturbances in different ways $\Longrightarrow$ heterogeneity in climate change sensitivity constitutes an important and underexplored dimension of regional inequality (Warsame et al., 2021; Acacha-Acakpo et al., 2024)

Contribution

- This paper contributes to the literature on the effects of institutional quality and climate change on regional economic growth
- We offer new empirical evidence, studying the context of a developing country where there are significant regional institutional disparities and a high degree of economic dependence on sectors vulnerable to climate change, such as agriculture and tourism.
- Using data at the regional level for Peru on a balanced panel of 24 departments over the period 2004 to 2019, we estimate spatial growth models that incorporate the efficiency of public resource allocation, and temperature and precipitation, as measures of climate change

Relevant literature

- The literature has recognized that institutional factors and environmental conditions are important factors explaining the regional economic growth (Baarsch et al., 2020; Aristizábal & García, 2021; Kouadio & Gakpa, 2022)

- In developing countries these factors are more significant, as regional inequalities are persistent and this is exacerbated by institutional weaknesses and climate vulnerability (Iammarino et al., 2019; Ibourk & Elouaourti, 2023; Paglialunga et al., 2023)

- Institutions play a critical role in economic growth (Acemoglu & Robinson, 2006):
  - Europe and North America: strong and transparent institutions correlate with higher growth rates and lower regional inequalities (Rodríguez-Pose & Di Cataldo, 2015; Charron et al., 2014)
  - Developing countries: institutional capacity tends to be weaker and more heterogeneous, resulting in uneven growth trajectories and regional economic divergence (Bosker & Garretsen, 2009)

- Climate change has emerged as a key determinant of regional economic performance, where countries heavily dependent on climate-sensitive sectors, such as agriculture and tourism, face greater economic risks $\Longrightarrow$ developing countries, but the evidence in these countries remains scarce (Burke et al., 2015; Damania et al., 2020; Baarsch et al., 2020)

This paper aims to offer new evidence on how institutional efficiency moderates the economic impact of climate variability shocks on regional growth

Data and descriptive evidence

Peru

- Population: 33 millions (Lima: 10.4 mill)
- Area: 1.1 million km2
- Tourism sector: 8% of GDP (the biggest sector); Agriculture sector: 5% of GDP

Figure 1. Peru



Table 1. Economic and social indicators in Peru and other countries

Country	GDP per capita (PPP)	GDP growth (annual, %)	Population (thousands)	Gini coef.	Poverty	Government effectiveness (WGI)
Peru	17,011	-0.4	33,846	40.7	5.9	-0.49
Colombia	21,246	0.7	52,321	53.9	7.7	-0.08
Ecuador	15,919	2.0	17,980	44.6	4.7	-0.49
Brazil	21,176	3.2	211,141	51.6	3.8	-0.55
Chile	33,145	0.5	19,659	43.0	0.5	0.72
Argentina	30,221	-1.9	45,538	42.4	1.2	-0.38
Mexico	25,243	3.4	129,740	43.5	2.3	-0.20
US	82,305	2.9	336,806	41.8	1.2	1.22
UK	59,911	0.3	68,492	32.4	0.5	1.16

Notes: Poverty: % below $2.15/day, 2017 PPP. Sources: World Bank Open Data. Most recent available year per indicator, 2018-2023. Government effectiveness from Worldwide Governance Indicators (WGI): ranging -2.5 to +2.5, where higher values denote better performance.

- Low economic growth
- high poverty levels
- weaker perceptions of institutional performance

Data and descriptive evidence

Figure 2. Departments of Peru



- Spatial analysis unit: 24 departments
- Departments are country subdivisions similar to US states and are granted a certain degree of autonomy
- The central region comprises the departments with the largest populations and highest levels of economic activity, including Ancásh, Huánuco, Lima, Pasco, Junín, Huancavelica, and Ica, accounting for 50% of the national population

Data and descriptive evidence

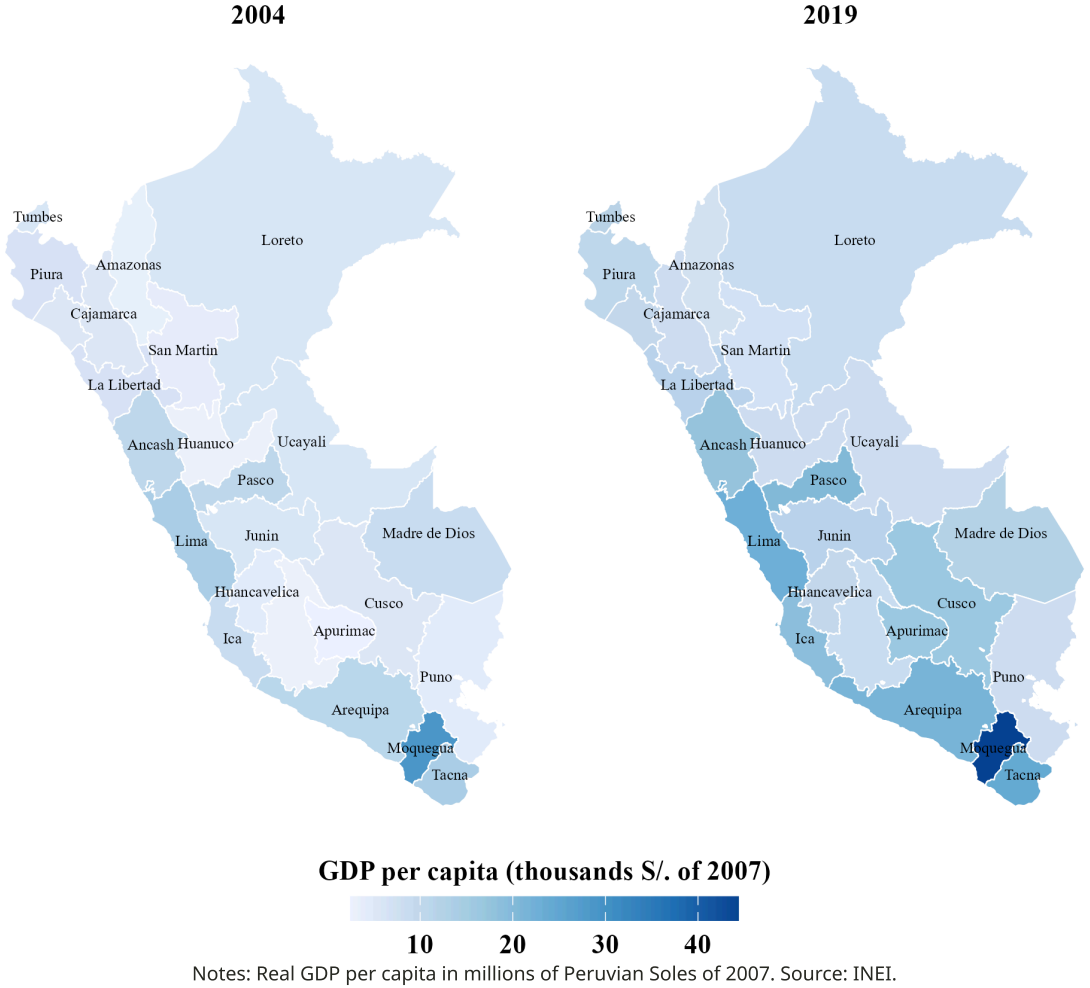
We uses a balanced panel dataset covering 24 departments of Peru over the period 2004 to 2019 (T=16), yielding 384 observations

Table 2. Variable description

Variable	Description	Source
Dependent variable		
Real GDP per capita	Gross domestic product per capita in thousands of Peruvian Soles at constant 2007	INEI
Independent variable of interest		
Institutional quality variables		
Efficiency index	Index efficiency in public resource allocation	RENAMU
Capacity index	Index capacity to provide public services and execute the policies	
Quality of public administration index (IQPA)	The average of the efficiency and capacity indexes	
Climate change variables		
Precipitation	Total annual precipitation in millimeters	Climate Change Knowledge Portal of the World Bank
Temperature	Mean annual temperature in C°	
Control variables		
% secondary	Percentage of the population with secondary education	ENAHO
Access to fixed internet	Percentage of households with internet access	ENAHO
Population density	Number of people per square km (in thousand)	INEI

Data and descriptive evidence

Figure 3. Real GDP per capita by departments



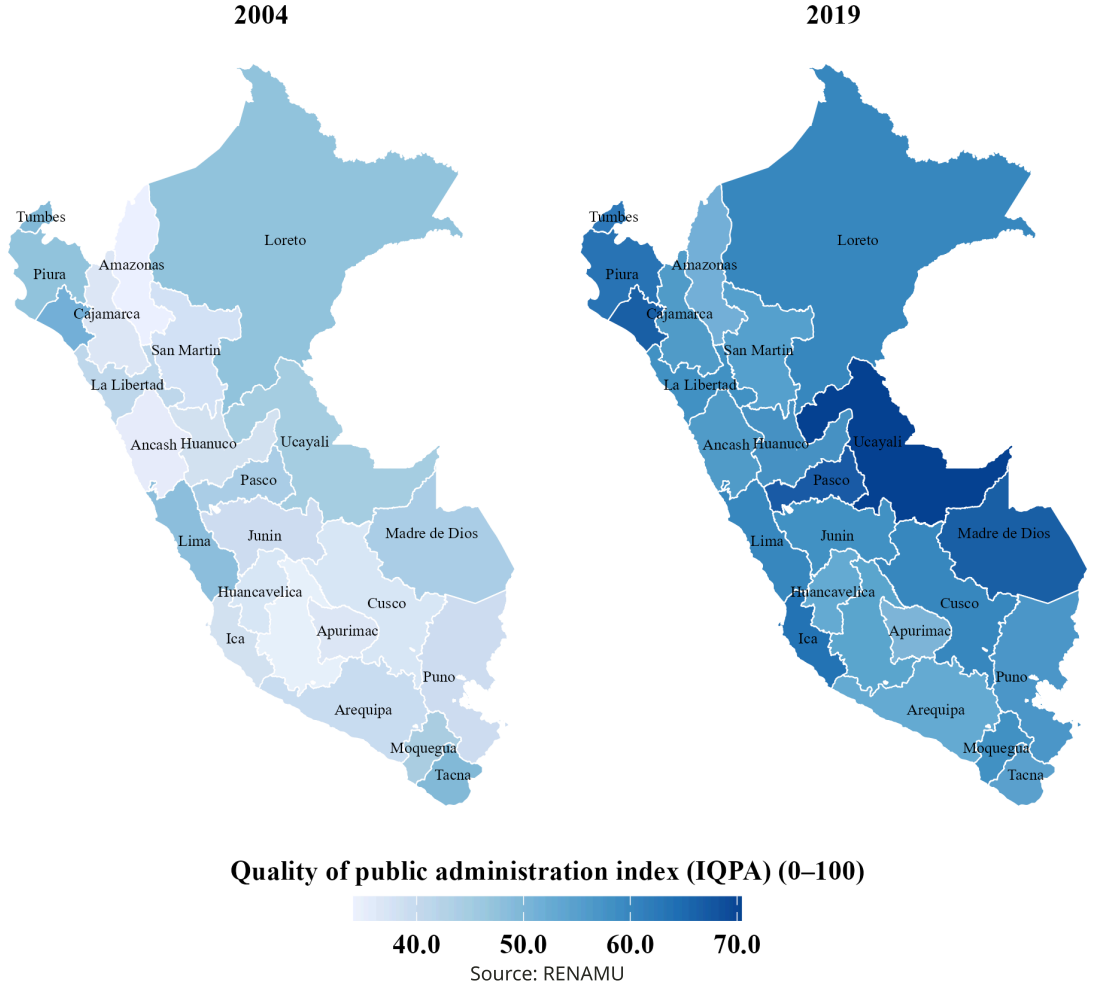
Notes: Real GDP per capita in millions of Peruvian Soles of 2007. Source: INEI.

There is a significant regional differences in economic performance, which have become more pronounced over time

Coastal departments, such as Ancash, Lima, Ica, Arequipa, Moquegua, and Tacna, present higher GDP per capita than those located in the Amazon region (Loreto, Amazonas, Ucayali, Madre de Dios, and Puno)

Data and descriptive evidence

Figure 4. Quality of public administration index (IQPA) by departments



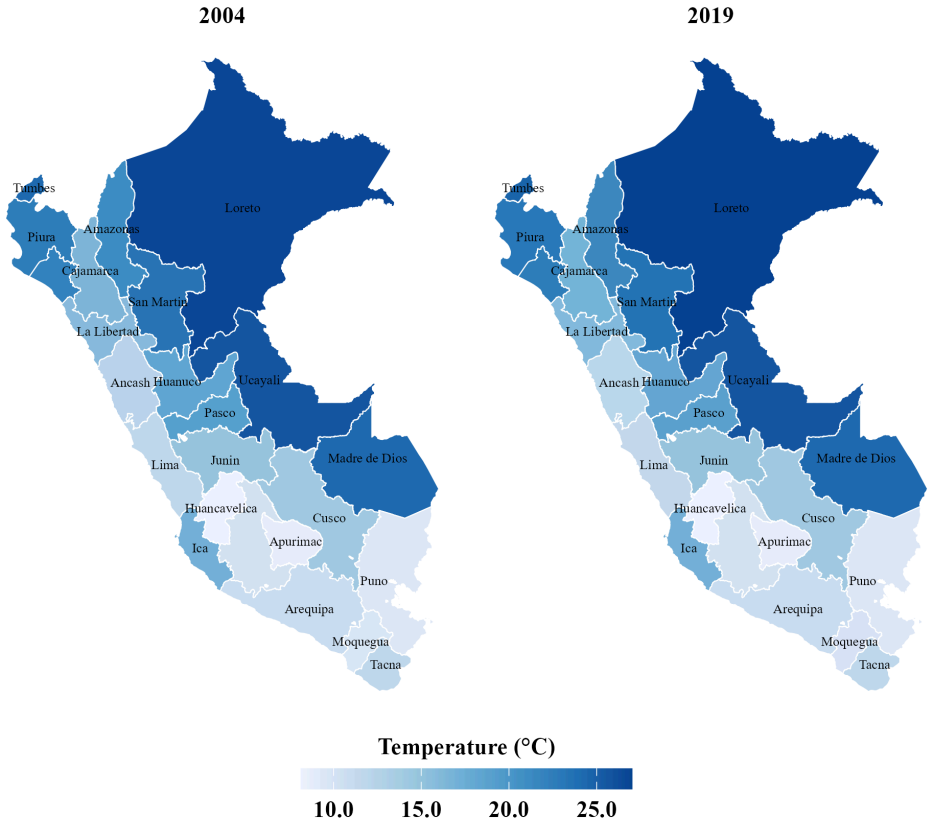
Source: RENAMU

High institutional quality: Ucayali, Pasco, Lambayeque, and Madre de Dios

Low institutional quality: Apurímac, Amazonas, Arequipa, and Huancavelica

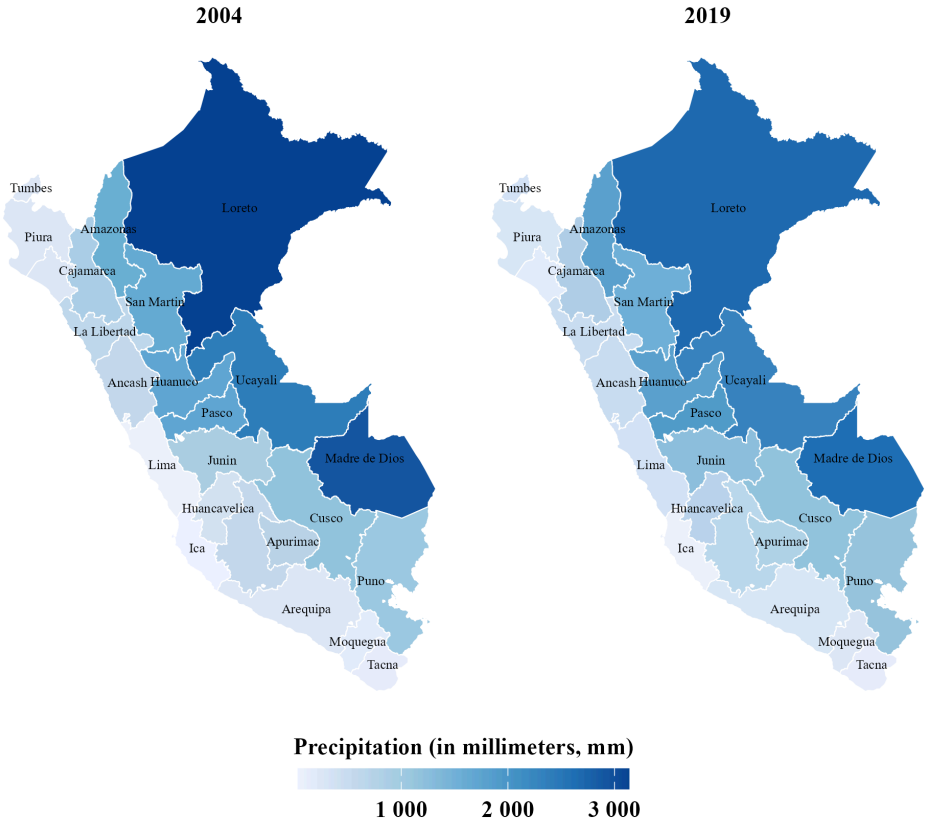
Data and descriptive evidence

Figure 5. Mean annual temperature



Source: Climate Change Knowledge Portal - World Bank

Figure 6. Total annual precipitation



Source: Climate Change Knowledge Portal - World Bank

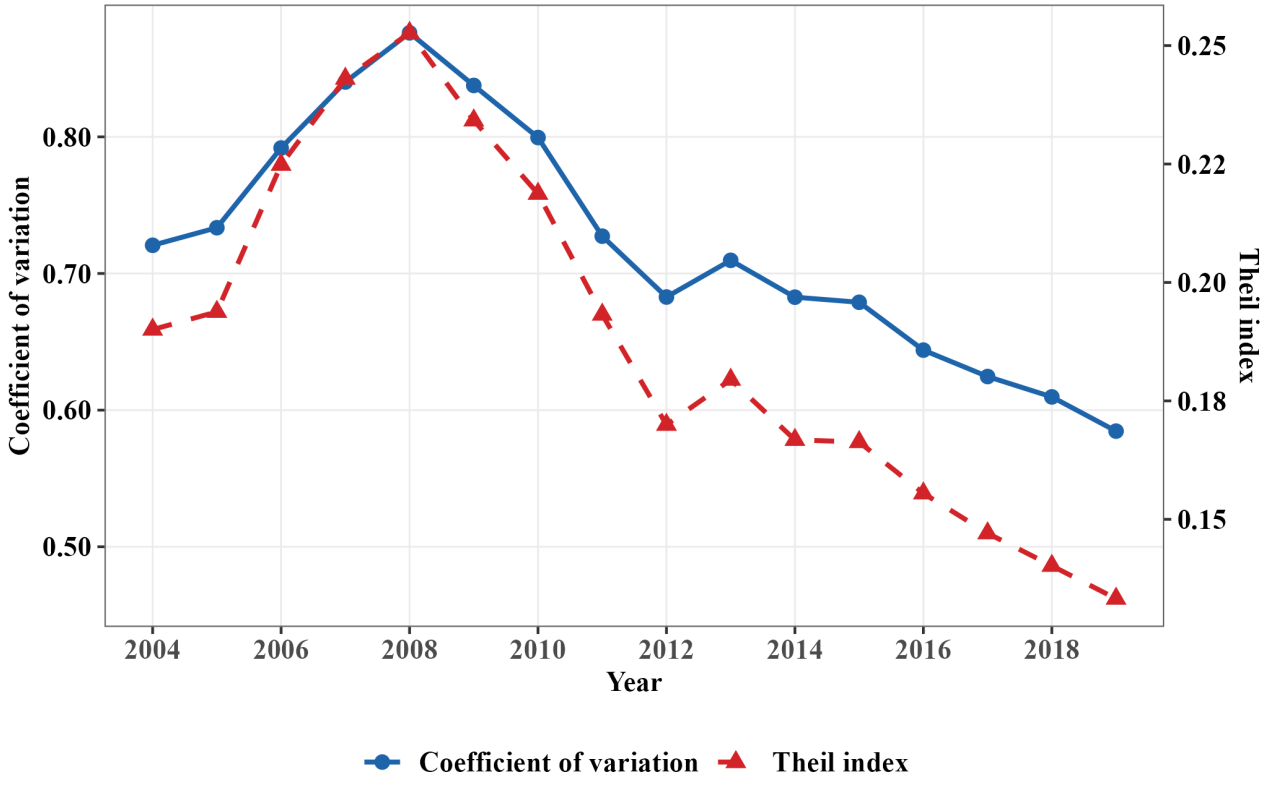
The highest average temperatures (>24°C) and accumulated precipitation (>2300 mm) were recorded in the Amazon region: Loreto, Ucayali, and Madre de Dios

Data and descriptive evidence

We tested the existence of sigma-type convergence in real GDP per capita:

- Sigma convergence refers to the reduction over time in the cross-sectional dispersion of income per capita across regions
- This measure captures the tendency of poorer regions to grow faster than richer ones

Figure 6. Coefficient of variation and Theil index of real GDP per capita



Econometric model

Beta convergence (Barro and Sala-i-Martin, 1991):

The functional form of the conditional Beta convergence models using panel data is as follows:

$$ln\left(\frac{Y_{it}}{Y_{it-1}}\right) = \alpha_i + \beta lnY_{it-1}$$
$$+\lambda InstQuality_{ite} + \delta lnY_{it-1} \times InstQuality$$
$$+\mu ClimateChange_{it} + \omega ClimateChange_{it}^2$$
$$+\theta X_{it} + u_{it}$$

Dependent variable: GDP per capita growth rate for department i between period t and period $t-1$

Y_{it-1}: the GDP per capita for department i in period $t-1$

β: convergence coefficient to estimate and is expected to be negative to reflect departments catching up to their steady-state levels

$InstQuality$: efficiency, capacity and IQPA

δ: measures whether institutional performance facilitates regional economic growth and convergence

$ClimateChange$: temperature and precipitation (quadratic term allows us to identify the nonlinear relationship, threshold effect)

X: is a matrix of explanatory variables representing different departmental steady states

α_i: fixed effects for each department and captures unobserved heterogeneity related to the steady state of each department

u_{it}: error term

Econometric model

In addition, we estimate the models of economic growth augmented by spatial effects

Table 3. Spatial panel data models

Spatial autoregressive model (SAR)	$\ln\left(\frac{Y_{it}}{Y_{it-1}}\right) = \alpha_i + \beta \ln(Y_{it-1}) + \rho W \ln\left(\frac{Y_{it}}{Y_{it-1}}\right) + \theta X_{it} + \varepsilon_{it}$
Spatial error model (SEM)	$\ln\left(\frac{Y_{it}}{Y_{it-1}}\right) = \alpha_i + \beta \ln(Y_{it-1}) + \theta X_{it} + \varepsilon_{it}$ $\varepsilon_{it} = \lambda W \varepsilon_{it} + u_{it}$
SAC model	$\ln\left(\frac{Y_{it}}{Y_{it-1}}\right) = \alpha_i + \beta \ln(Y_{it-1}) + \rho W \ln\left(\frac{Y_{it}}{Y_{it-1}}\right) + \theta X_{it} + \varepsilon_{it}$ $\varepsilon_{it} = \lambda W \varepsilon_{it} + u_{it}$
Spatial Durbin model (SDM)	$\ln\left(\frac{Y_{it}}{Y_{it-1}}\right) = \alpha_i + \beta \ln(Y_{it-1}) + \rho W \ln\left(\frac{Y_{it}}{Y_{it-1}}\right) + \theta X_{it} + \gamma W X_{it} + \varepsilon_{it}$

W is the standardized queen-type first-order spatial contiguity matrix for all the models

Results

Table 4. Estimates of spatial panel data models of growth

Y = GDP per capita growth rate

	OLS	SAR	SEM	SAC	SDM
ln GDP per capita (t-1)	-0.103***	-0.093***	-0.097***	-0.090***	-0.134***
	(0.0294)	(0.0277)	(0.0279)	(0.0283)	(0.0306)
IQPA	0.061***	0.063***	0.061***	0.060***	0.059***
	(0.0170)	(0.0160)	(0.0159)	(0.0162)	(0.0162)
ln GDP per capita (t-1)*IQPA	-0.027***	-0.028***	-0.028***	-0.028***	-0.028***
	(0.0067)	(0.0063)	(0.0063)	(0.0063)	(0.0064)
Precipitation	0.004	0.001	0.001	0.005	0.006
	(0.0005)	(0.0609)	(0.0001)	(0.0005)	(0.0005)
Precipitation^2	-0.008	-0.001	-0.001	-0.001	-0.009
	(0.0009)	(0.0001)	(0.001)	(0.0002)	(0.0001)
Temperature	0.0958**	0.091**	0.102**	0.105**	0.116
	(0.0383)	(0.0361)	(0.0415)	(0.0442)	(0.1011)
Temperature^2	-0.002**	-0.002**	-0.002*	-0.002*	-0.004
	(0.0010)	(0.0010)	(0.0011)	(0.0011)	(0.0017)
λ		0.198***		-0.086	0.223***
		(0.0628)		(0.2114)	(0.0633)
ρ			0.238***	0.313*	
			(0.0637)	(0.1839)	
Speed of convergence (%)	6.08	5.69	5.85	5.57	7.15
Half-life (years)	11.38	12.16	11.83	12.43	9.68
LM Lag	7.155***				
LM Error	0.102				
Robust LM Lag	8.238***				
Robust LM Error	1.186				
N	384	384	384	384	384

Notes: Standard errors in parenthesis. Speed of convergence = $\ln(1-T\beta)/T$, Half-life = $\ln(2)/speed\ of\ convergence$
***p<0.01, **p<0.05, *p<0.1

Results consistently confirm the presence of conditional β-convergence among Peruvian departments $\Longrightarrow$ poorer regions tend to grow faster than richer ones

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- Regarding the effect of the institutional variable, we observe that it is positive and statistically significant $\Longrightarrow$ better public administration performance positively impacts the region's economic growth
- The interaction term between institutional performance and regional economic growth show that good institutional performance through institutional efficiency and capacity helps facilitate regional economic convergence in Peru

Results

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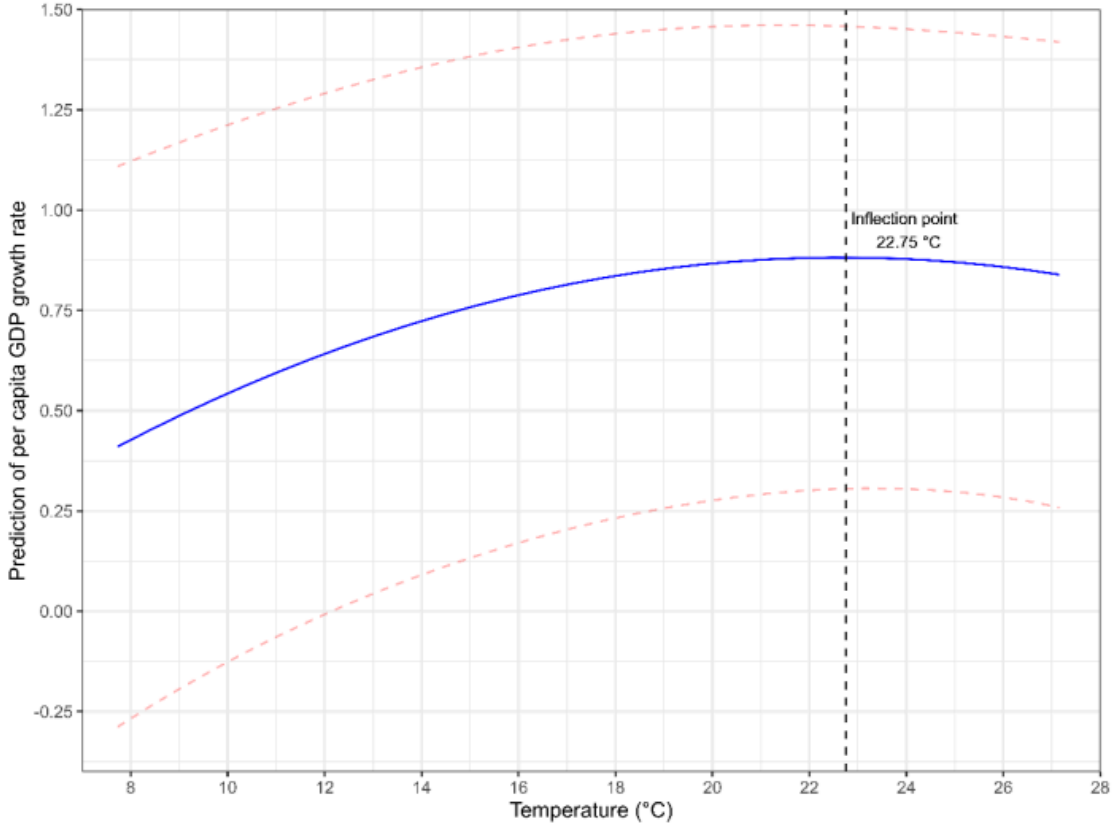
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Climate change impacts are affecting the economic growth in Peru during the period analyzed

Results

Figure 7. Effect of temperature on economic growth



Notes: Prediction of per capita GDP growth rate with variables in its means for different temperature levels (blue line) with 90% confidence interval (red lines). Vertical line represents the inflection point calculated as $\frac{-\hat{\beta}_{Temperature}}{2\hat{\beta}_{Temperature^2}}$, which results from deriving the economic growth equation with respect to temperature and equals zero and isolating the temperature.

There is a positive effect of temperature on economic growth, but this effect is negative after 22.75°C

Conclusions

- In this paper we analyze the effects of institutional quality and climate change on economic growth at the intra-regional level in the context of a developing Latin American country, Peru
- This study makes a novel contribution in its simultaneous incorporation of institutional and climate risk variables and spatial dependence into a panel model of intra-regional growth, which has been little explored in the literature on regional growth in developing countries
- The findings highlight the critical role of local institutional quality in fostering economic growth
- This study shows that climate change has a complex relationship with regional economic growth
- Temperature significantly impact Peru's economic growth over the analyzed period. Including a quadratic term for temperature allows for identifying a nonlinear relationship consistent with economic theory, suggesting that rising temperatures harm growth beyond a certain threshold
- Climate change poses a serious challenge to regional growth. Addressing its adverse effects requires adaptive strategies, such as investments in climate-resilient infrastructure and sustainable agricultural practices, to mitigate temperature-related economic losses